Why netlists

Every SPICE solve starts from a netlist , nodes, elements, and a ground reference

The idea

- GOLDEN: 5 elements, 2 unknown nodes (1, 2), ground 0

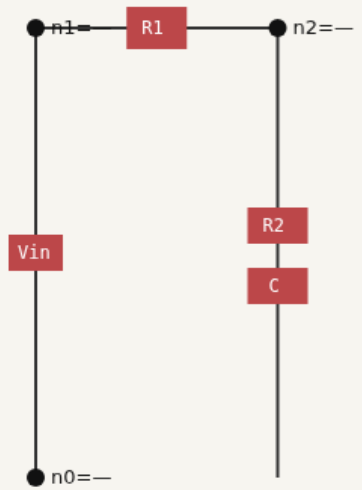
Netlist

NETLIST NODES

Netlist

Step 1 / 5 · nl · module01-01-netlist-nodes

5 elements



Explanation

Toy netlist with five elements.

- JSON

```
n=5
```

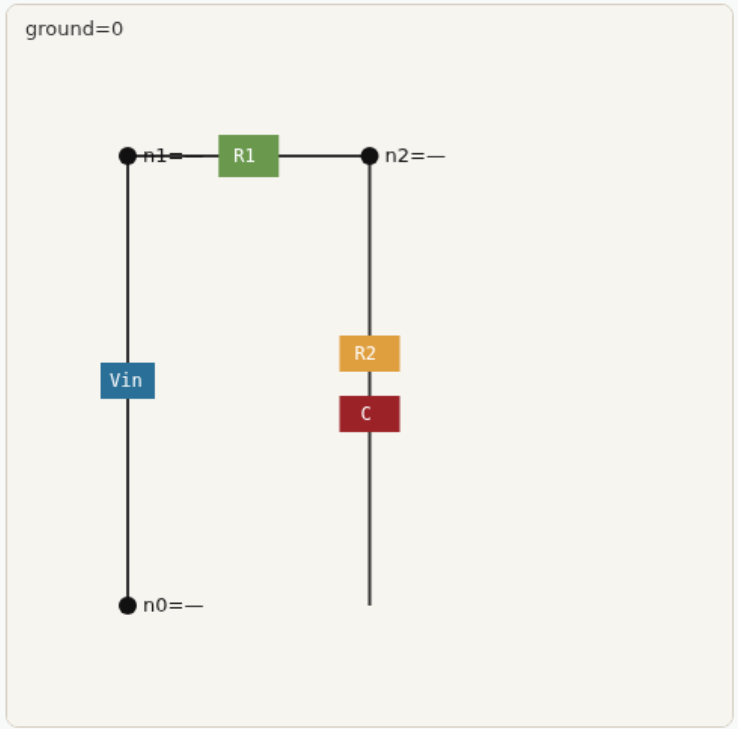
Netlist

Ground

NETLIST NODES

Ground

Step 2 / 5 · gnd · module01-01-netlist-nodes



Explanation

Node 0 is ground reference.

- gnd

```
0
```

Ground

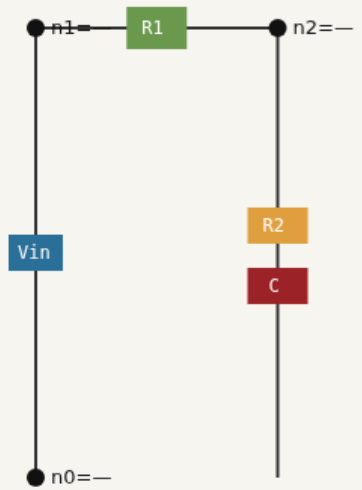
Unknowns

NETLIST NODES

Unknowns

Step 3 / 5 · un · module01-01-netlist-nodes

unknowns=2



Explanation

Unknown voltages: nodes 1 and 2.

- MNA

```
2
```

Unknowns

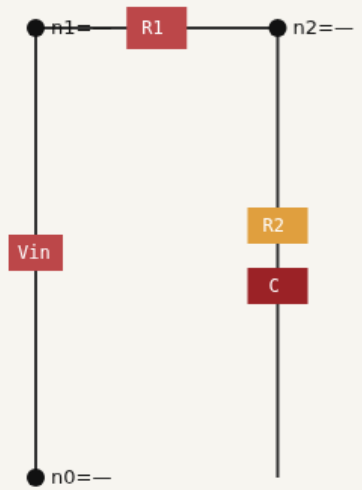
Elements

NETLIST NODES

Elements

Step 4 / 5 · el · module01-01-netlist-nodes

types



Explanation

V, R, R, C, D types.

- types

```
V/R/C/D
```

Elements

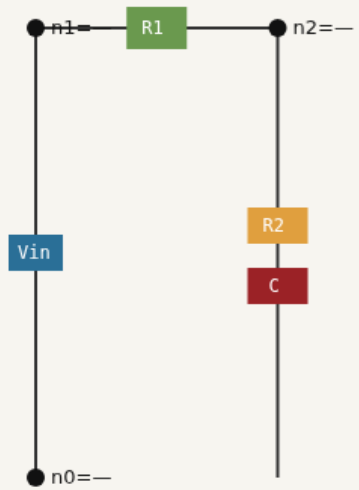
Count

NETLIST NODES

Count

Step 5 / 5 · cnt · module01-01-netlist-nodes

{"elementCount":5,"nodeCount":3,"unknownNodes":2,"ground":"0","ids



Explanation

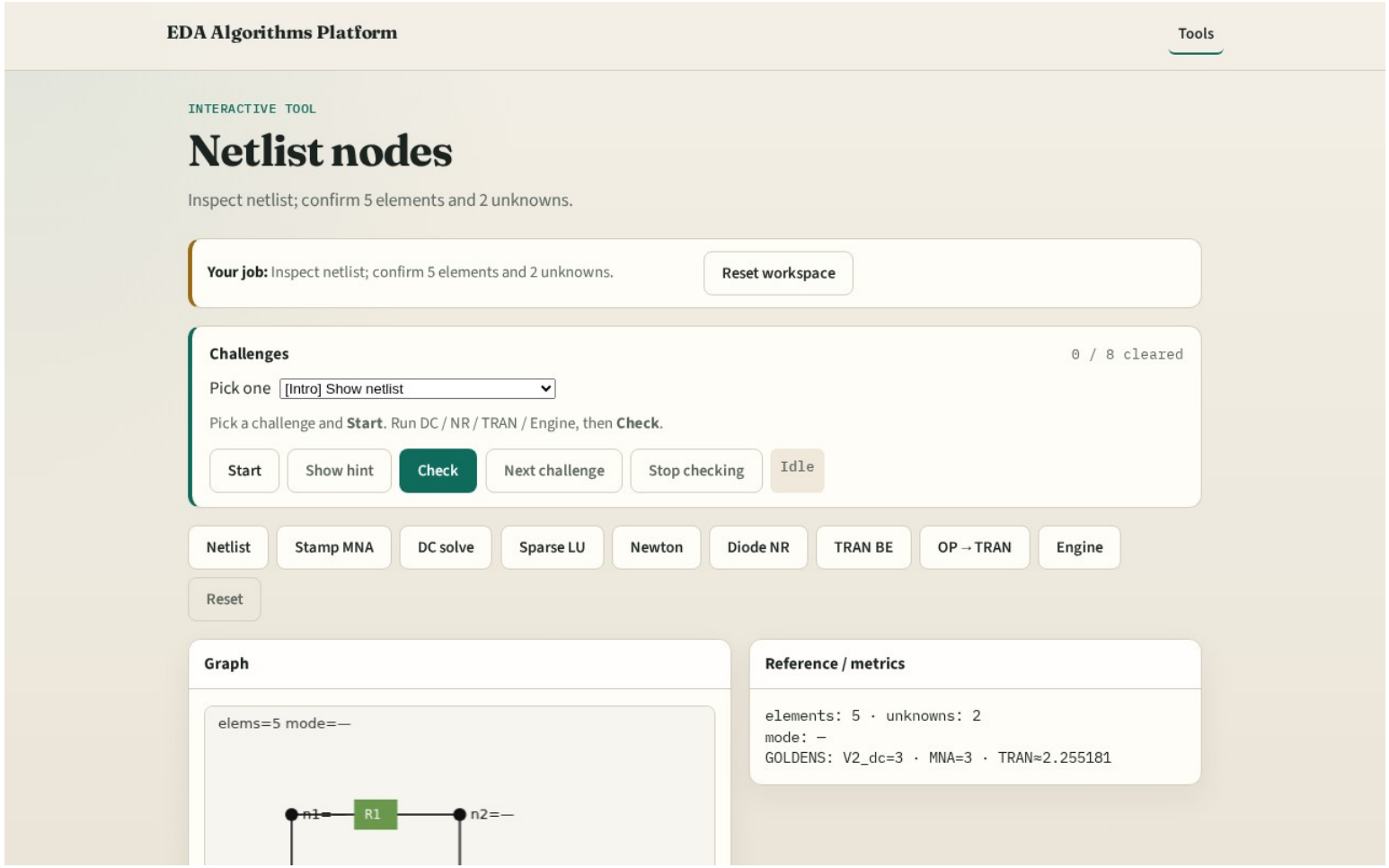
GOLDEN element_count=5.

- GOLDEN

```
5
```

Count

Browser lab track



Browser lab starter

Implement track

- Call netlist_stats(); assert element_count==5, unknown_nodes==2, ground=='0'

Pitfalls

- Counting ground as unknown
- Ignoring element ids vs node names

Your turn

- Next: stamp R and V into MNA